

A counterexample to Takao's self-duality question

Alec Kriebel

Independent researcher

ORCID: [0009-0001-9320-500X](https://orcid.org/0009-0001-9320-500X)

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Abstract

Takao asked whether a finite flat commutative group scheme over a Witt ring, killed by p and having quasi-supersingular special fiber, must be Cartier self-dual in ranks p^{2n} with $n \geq 3$. For every $p > 3$ and $n \geq 3$, we give a counterexample over $W(\mathbb{F}_p)$. A six-dimensional Dieudonné module has an explicit filtration by supersingular elliptic p -torsion modules, an explicit finite Honda complement, and an image-intersection invariant distinguishing it from its dual. Both the special fiber and its finite flat Witt-ring lift are non-self-dual. The construction is an application of classical Dieudonné and finite Honda theory.

1 The question and result

Let $p > 3$ be prime. A finite commutative group scheme H over a perfect field of characteristic p , killed by p , is *quasi-supersingular* if it has a subgroup filtration whose successive quotients are $E_i[p]$ for supersingular elliptic curves E_i over that field. This is Takao's Definition 1(3) [5]. His Proposition 2 states that, for a finite flat commutative $W(k)$ -group scheme $\mathcal{G}$ killed by p , of rank p^{2n} , a quasi-supersingular special fiber forces both the special fiber and $\mathcal{G}$ to be self-dual when $n \leq 2$. The unnumbered question immediately following that proposition asks whether the assertion remains true for $n \geq 3$ [5, p. 2479].

Theorem 1. *For every prime $p > 3$ and integer $n \geq 3$, put $k = \mathbb{F}_p$ and $W = W(k)$. There is a finite flat commutative group scheme $\mathcal{G}_n/W$, killed by p and of rank p^{2n} , whose special fiber H_n is quasi-supersingular, such that*

$$H_n \not\cong H_n^{\mathrm{D}}, \quad \mathcal{G}_n \not\cong \mathcal{G}_n^{\mathrm{D}}.$$

Here $^{\mathrm{D}}$ denotes Cartier duality.

Thus the answer to the stated higher-rank question is negative, with the first possible counterexample at $n = 3$, using Takao's affirmative result for $n \leq 2$. The hypothesis concerns the special fiber: we do not require its filtration to lift to W .

The module used below belongs to the classical cyclic-word framework. In the convention of Pries–Ulmer it is $M(vfvff)$ [4, §§3.2.1, 4.1]; in the forward-arrow order of our contravariant presentation its string is $ffvfvv$. The module construction and its duality classification are prior mathematics. Oort's cyclic integral construction also supplies a balanced supersingular completion [3, §§2.5–2.6]. We give direct calculations of the filtration, duality obstruction, and finite Honda complement to answer the specific question without relying on the cyclic-word classification. No first-discovery or first-priority claim is made. The supporting package records the bounded literature audit and its source-access and version limits.

2 The six-dimensional construction

We use contravariant Dieudonné theory over k as in [2, §2]. Write σ for Frobenius on k and also for its lift to W . On a module killed by p , F is σ -semilinear, V is σ^{-1} -semilinear, and $FV = VF = 0$. Let M have

basis $e_0, \dots, e_5$, with operators specified by

$$\begin{array}{c|cccccc} & e_0 & e_1 & e_2 & e_3 & e_4 & e_5 \\ \hline F & e_1 & e_2 & 0 & e_4 & 0 & 0 \\ V & e_5 & 0 & 0 & e_2 & 0 & e_4 \end{array}$$

and extended with the indicated semilinearity. All displayed scalars are in $\mathbb{F}_p$. Direct computation gives

$$\begin{aligned} FV &= VF = 0, & F^3 &= V^3 = 0, \\ \text{im } F &= \ker V = \text{span}_k(e_1, e_2, e_4), & \text{im } V &= \ker F = \text{span}_k(e_2, e_4, e_5). \end{aligned}$$

The finite Dieudonné anti-equivalence produces a commutative k -group scheme H killed by p , of rank p^6 [2, Proposition 2.5].

2.1 Quasi-supersingularity

Let I denote the two-dimensional module on x, y with

$$Fx = Vx = y, \quad Fy = Vy = 0.$$

Over algebraically closed k , I is the module of the p -torsion of a supersingular elliptic curve. This is the rank-two case of [2, Lemma 4.9]: the module is deformable and $\text{im } F = \text{im } V$.

Set

$$u = e_1 + e_3 + e_5, \quad v = e_2 + e_4, \quad w = 2e_1 + e_3, \quad z = e_2, \quad a = e_0, \quad b = e_1.$$

These form a basis. Indeed,

$$e_0 = a, \quad e_1 = b, \quad e_2 = z, \quad e_3 = w - 2b, \quad e_4 = v - z, \quad e_5 = u - w + b.$$

In this basis the nonzero images are exactly

$$\begin{aligned} Fu &= v, & Fw &= v + z, & Fa &= b, & Fb &= z, \\ Vu &= v, & Vw &= z, & Va &= u - w + b. \end{aligned}$$

Consequently

$$0 = M_0 \subset M_1 = \text{span}_k(u, v) \subset M_2 = \text{span}_k(u, v, w, z) \subset M_3 = M$$

is an F, V -stable flag. The three quotient modules are I , using, respectively, the pairs (u, v) , (w, z) , and (a, b) . Exact contravariance gives subgroups $H_i \subset H$ with modules M/M_{3-i} , for $0 \leq i \leq 3$. Their successive quotients correspond in reverse order to the three module factors. Each is supersingular elliptic p -torsion; hence H is quasi-supersingular.

2.2 Failure of Cartier self-duality

For a finite Dieudonné module T over perfect k , define

$$\delta(T) = \dim_k(\text{im } F_T^2 \cap \text{im } V_T^2).$$

The images are k -subspaces because the field is perfect. An isomorphism commuting with F, V carries both images to the corresponding images, so δ is an isomorphism invariant. For M ,

$$\text{im } F^2 = ke_2, \quad \text{im } V^2 = ke_4, \quad \delta(M) = 0.$$

On $M^D = \text{Hom}_k(M, k)$ the dual operators are

$$(F^D\phi)(t) = \sigma(\phi(Vt)), \quad (V^D\phi)(t) = \sigma^{-1}(\phi(Ft)) \quad [2, \text{Definition 2.3}].$$

Since our basis matrices have coefficients in $\mathbb{F}_p$, their dual matrices are V^t and F^t with the appropriate semilinearity. Thus $(F^D)^2(e_4^*) = e_0^*$ and $(V^D)^2(e_2^*) = e_0^*$, and all other basis images of these second iterates vanish. It follows that

$$\text{im}(F^D)^2 = \text{im}(V^D)^2 = ke_0^*, \quad \delta(M^D) = 1.$$

Hence $M \not\cong M^D$, and compatibility of Dieudonné theory with Cartier duality gives $H \not\cong H^D$. The transpose description here uses the prime-field coefficients; the displayed semilinear dual formulas are the general convention.

3 An actual finite flat lift and all higher ranks

View M as a W -module through $W \rightarrow k$ and put

$$L = \text{span}_k(e_0, e_3, e_5) \subset M.$$

We verify the finite Honda hypotheses explicitly. The W -length of M is six and $pM = 0$. The semilinear operators satisfy $FV = VF = p$ on M . The restriction of V to L sends its displayed basis to e_5, e_2, e_4 and is injective, since k is perfect. Finally, L complements $\text{im } F = \text{span}_k(e_1, e_2, e_4)$, so the natural map

$$L/pL \longrightarrow M/\text{im } F$$

is an isomorphism. These are exactly the finite Honda conditions of Fontaine–Laffaille [1, §9.4], in the p -torsion formulation of Hoshi [2, Remark 3.5.1]. The exactness identities above also verify Hoshi’s deformability condition.

Since $p \neq 2$, Hoshi’s anti-equivalence [2, Proposition 3.11(2)] produces a finite flat commutative W -group scheme $\mathcal{G}$, killed by p , from this finite Honda system. Its special fiber has module M , by [2, Proposition 3.11(5)], and is therefore H . Its rank is p^6 : finite flat rank is constant over the local base and equals the special-fiber rank. Cartier duality commutes with base change, so an isomorphism $\mathcal{G} \simeq \mathcal{G}^D$ would imply $H \simeq H^D$. Thus $\mathcal{G}$ also fails self-duality. This is a finite Honda realization over W ; an integral Dieudonné lattice for a group over k would not, on its own, establish this mixed-characteristic assertion.

For $n \geq 3$, equip I with the finite Honda subspace kx and set

$$M_n = M \oplus I^{\oplus(n-3)}, \quad L_n = L \oplus (kx)^{\oplus(n-3)}.$$

The same Honda conditions hold on each block. They give $\mathcal{G}_n/W$ of rank p^{2n} . Its special fiber is $H \times E[p]^{n-3}$ and has a quasi-supersingular filtration by extending the filtration already constructed with the product elliptic factors. Since $F^2 = V^2 = 0$ on I and on its dual,

$$\delta(M_n) = 0, \quad \delta(M_n^D) = 1.$$

Therefore H_n and, by base change, $\mathcal{G}_n$ are not Cartier self-dual. This proves Theorem 1.

Scope and verification. These examples over $\overline{\mathbb{F}}_p$ disprove the universal assertion for perfect residue fields; no descent to every perfect field is asserted. The result concerns general finite group schemes and does not produce a principally polarized Jacobian or decide the surrounding Coleman conjecture. The proof is symbolic for every allowed p, n ; the companion standard-library verification programs check the basis, operators, invariant, Honda conditions, direct sums, and nontrivial Frobenius twists over finite fields. Those computations corroborate the proof and do not replace the cited classification theorems or certify historical priority.

AI use and review status. AI tools were used extensively in solving, verification, literature research, writing, computation, and adversarial review. This note is an unrefereed preprint. The automated reviews are not independent external human peer review; no such human review is claimed.

References

- [1] J.-M. Fontaine and G. Laffaille, *Construction de représentations p -adiques*, Ann. Sci. École Norm. Sup. (4) **15** (1982), no. 4, 547–608, [Numdam original](#).
- [2] Y. Hoshi, *Pseudo-rigid p -Torsion Finite Flat Commutative Group Schemes*, March 2021 revised author manuscript, <https://www.kurims.kyoto-u.ac.jp/~yuichiro/rims1911revised.pdf>. The numbered statements cited are those of this manuscript. Published in J. Number Theory **229** (2021), 261–276, [doi:10.1016/j.jnt.2021.04.026](https://doi.org/10.1016/j.jnt.2021.04.026).

- [3] F. Oort, *Simple p -kernels of p -divisible groups*, June 2004/revised January 2005 author manuscript, <https://webpace.science.uu.nl/~oort0109/A-SimpleFinGrSch2.ps>. Published in Adv. Math. **198** (2005), 275–310, [doi:10.1016/j.aim.2005.03.015](https://doi.org/10.1016/j.aim.2005.03.015).
- [4] R. Pries and D. Ulmer, *On BT_1 group schemes and Fermat curves*, New York J. Math. **27** (2021), 705–739, <https://nyjm.albany.edu/j/2021/27-27v.pdf>.
- [5] N. Takao, *Quasi-supersingular finite flat commutative group schemes and the Coleman conjecture on torsion points on curves*, in *Arithmetic Homotopy and Galois Theory*, Oberwolfach Reports **20** (2023), no. 3, Report 42/2023, 2476–2480, [doi:10.4171/OWR/2023/42](https://doi.org/10.4171/OWR/2023/42).